VIQCR 1.1

The complete VEX IQ competitive rating algorithm

Cooperative performance. Solo skills. Partner-adjusted consistency.

A separate IQ model that evaluates shared-score contribution and repeatable own performance across different partners.

COMPONENT	WEIGHT
Teamwork contribution	55%
Driving skills	17.5%
Autonomous coding skills	17.5%
Partner-adjusted consistency	10%

MODEL STATUS

Executable candidate, scenario-tested. Historical IQ calibration and live website/database integration remain pending. This document publishes the complete algorithm specification; it does not deploy the model.

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Developed November 1, 2026. Model: `VIQCR-1.1-candidate.1`.

Status: executable candidate with automated scenario tests. Historical calibration, official-data import, IQ API/UI integration and deployment are not included. This document specifies the IQ algorithm; it does not change the existing V5 leaderboard. The 55/17.5/17.5/10 weights and all statistical constants below are project policies to validate, not VEX rules.

The implementation is website/lib/viqcr.mjs, shared through algorithm/viqcr.mjs. Tests are website/tests/viqcr.test.mjs.

Purpose and outputs

Estimate a team's current ability to contribute to cooperative matches, perform solo driving tasks, and execute autonomous coding skills. Publish four component ratings, a combined conservative score, uncertainty, provisional status, evidence coverage and an event/session audit. Keep official competition standings and honors alongside the estimate.

```
Combined strength = 0.55 × Teamwork
                  + 0.175 × Driving Skills
                  + 0.175 × Autonomous Coding Skills
                  + 0.10 × Partner-adjusted Consistency
```

There is no opposing IQ alliance, defense rating, win/loss Elo, autonomous bonus, autonomous win point or V5 elimination advancement term. A station labeled red or blue identifies one of the partners. V5 scores and IQ scores must never enter the same model or leaderboard.

The rating measures observable competitive performance. It cannot measure the quality of a student's notebook, interview, design process, conduct or learning from a match score. Those achievements remain visible through official records, rather than an invented numeric proxy.

Verified rules and competition coverage

The rules baseline reviewed is Level Up v2.1, dated October 8, 2026. The official Q&A is also binding. Importers must retain the applicable manual/Q&A revision with each dataset; later revisions require review before rebuilding.

Two robots share a sixty-second teamwork challenge. Solo skills have driving and autonomous coding modes. Level Up awards different values for goal levels. These game details come from the official current-game overview. The model consumes referee-confirmed points, so it does not guess scoring legality from object counts or reuse another season's point values.

Official event rules use qualification averages with prescribed drops, protected no-shows and extra-match exclusions. League participation has a 60% threshold. Finals partners follow consecutive qualification ranks; finals are score-based, with special ties and an overall final after division finals. Skills opportunities are three per mode/session. Official skills totals use mode bests from one event/session; event and global tie orders differ. See the tournament rules, T8-T19.

Competition aspect	Algorithm treatment	Official record/display responsibility
Match consistency	Rolling partner-adjusted own contribution, variation and coverage	Show sample size, partner diversity and uncertainty; do not label a team as carried
Teamwork qualifications	Joint estimate from one shared score and both partner identities	Import team-specific official averages, rank and participation
Dropped qualification scores	Retain valid played scores as ability evidence	`summarizeIQQualifications` applies the published drop counts; official order resolves ties
Extra qualification appearances	Can inform ability within the same capped evidence budget	Exclude the flagged team's extra appearance from official average/participation
Finals	Shared-score evidence with a small total budget	Keep seed, division, match stop time, tiebreak series, final rank and verified champion
Multiple divisions and overall finals	Ingest each played match once, including overall finals	Division champion and overall champion are distinct fields
Driving skills	Independent solo estimate from peak and repeatability	Show official peak, attempts when known, stop time and rank
Autonomous coding skills	Independent solo estimate with equal skills weight	Show official coding peak and rank; do not invent a teamwork auto period
League ranking/finals sessions	One dated update per verified session	Use the same session for official combined skills; preserve league participation standings
Skills-only events	Update skills; teamwork stays unobserved	Clearly label event format and missing teamwork coverage
Elementary and middle school	Same equations; separate grade filters	Import verified season-specific classification; keep unknown explicit
Mixed-grade events	Model the actual partners without a grade handicap	Filter each team's verified grade after calculation
Regional, national, signature and Worlds events	Actual ability evidence; no prestige multiplier	Display official level and scope without inferring it from names
Judging, notebooks, interviews, design and STEM learning	No match-derived numerical proxy	Preserve official judged awards and publicly released judging evidence
Sportsmanship, student-centeredness, safety and robot inspection	Respect official eligibility/rulings	Report verified sanctions/eligibility, never speculate about compliance

Competition aspect	Algorithm treatment	Official record/display responsibility
Driver switch, loading, resets and interaction rules	Their legal effects are already reflected in official scores	Preserve official rulings; no guessed violations or per-driver attribution
No-show, robot unavailable or teamwork DQ	Exclude affected shared score from ability fitting; audit reason	Keep team-specific awarded scores and sanctions in official standings
Skills DQ	Official zero is a scored attempt	Retain the DQ flag and official zero
Practice, void, unplayed and replaced matches	Exclude from model	Keep audit identifiers; a replacement replay is distinct from a finals tiebreak
Rule/scoring changes between seasons	Separate profile and replay per IQ season	Keep game, manual version and provenance; never compare raw scores across games

Qualification DQs affect the penalized team's awarded score while its partner may retain points; finals DQs affect the alliance. Skills DQs are zeroes. Driver-switch and permitted robot-handling rules also affect real performance. These distinctions are established in GG6, GG10 and GG11. The candidate deliberately avoids interpreting a teamwork sanction as evidence that either robot physically scored zero.

Judged honors depend on criteria beyond results and are not uniformly offered. Preserve official Excellence, Design and other honors as separate achievements, using the Guide to Judging: Awards. No award grants VIQCR points or resolves a performance-rating tie. Qualification pathways and invitations are official administrative outcomes, not rating bonuses.

State, units and season normalization

Each team has four component means m and variances v. The three performance modes and the consistency snapshot each start at $m=0,\ v=1$, with no evidence. Internal rating points are $1500 + 200m$. Means are standardized ability units, not raw match points or percentages.

Each IQ season has a frozen normalization profile:

```
{
  program: 'VIQRC', seasonId: 'official-iq-season-id-as-string',
  version: 'iq-season-calibration-v1', rulesVersion: 'Level Up v2.1 + reviewed Q&A',
  trainedThrough: '2026-09-01T00:00:00Z',
  teamwork: { center: 100, scale: 25 },
  driving: { center: 50, scale: 20 },
  coding: { center: 40, scale: 20 }
}
```

These numerical values are illustrative fixtures, not calibrated Level Up constants. The implementation deliberately ships no purported production IQ normalization or guessed official season ID.

Fit centers and scales using a documented initial calibration window in the same game, before the first rated event. Suggested estimator: median for center, and $\max(\text{positive minimum}, 1.4826 \times \text{MAD})$ for scale. Teamwork uses alliance-score observations; skills use team/session observations of the same peak/repeatability statistic used below. Equalize team/session representation and publish cohort size, date range, missingness and connectivity. Establish a positive fallback scale before using a low-scoring or all-zero coding cohort.

The calibration window trains the profile; it is not then replayed as rated evidence using a profile learned from its own outcomes. Start rated replay after `trainedThrough`. Freeze the profile for that model/season release. A revised profile requires a full rebuild and a new version; both version changes and silent parameter changes are rejected in an existing replay. Do not recenter scores on each event's own field: an elite event and a beginner event should not both be forced to average strength.

Use one global IQ-season profile for mixed-grade connectivity. Grade and regional leaderboards filter this same model. Common scales are a modeling choice, not proof that disconnected regions are perfectly comparable; sparse cross-region partnerships must be disclosed in validation.

Cooperative teamwork estimate

For a match with partner set `P` of size one or two:

```
baseline = teamwork.center × |P| / 2
expected points = baseline + teamwork.scale × Σ prior_mean_i
raw target = (official shared points - baseline) / teamwork.scale
fitted target = Σ prior_mean_i + clamp(raw target - Σ prior_mean_i, -3, +3)
```

Ordinary matches have two partners. The one-team case is reserved for an officially scheduled solo appearance, such as a finalist without a replacement partner. A normal two-team match with a no-show is excluded rather than reinterpreted as scheduled solo evidence.

Fit both teams jointly with Gaussian regularization toward their frozen priors. This is an additive contribution estimate with uncertainty, not a claim that each robot scored a known number of points. In standardized units:

```
y = m_i + m_j + noise
```

A known strong partner explains more of the shared score. When both robots have equal uncertainty, the unexplained residual divides equally; the more uncertain robot can absorb more of it when uncertainties differ. Skills are kept separate, so a high solo score is not assumed to prove the same teamwork contribution.

The implementation starts each session with diagonal prior covariance and performs a joint Gaussian update for every eligible observation, retaining the induced partner correlations during that fit:

```
x = indicator vector for the participating robots
R = 1 / observation_weight
K = P x / (R + x^T P x)
m_new = m + K(y - x^Tm)
P_new = P - (P x)(P x)^T / (R + x^T P x)
```

Targets are bounded against frozen priors, so reversing match input order produces the same posterior, within floating-point tolerance. Match timestamps do not silently turn the second partner into an opponent. The fitted component change is then capped at +/-0.6 ability units per session.

Evidence weights use the strongest applicable limit:

```
schedule limit = min over participating teams of min(1, 6 / appearances_this_session)
pair limit = min(1, 2 / appearances_of_this_exact_partner_set)
stage limit = 1 for qualifications
            = 0.35 / max(participating teams' finals appearances) for finals
observation weight = min(schedule limit, pair limit, stage limit) × row reliability
```

Reliability defaults to 1 for verified ordinary evidence; zero excludes the row. Same-partner repeats saturate at two units of information. All finals, division finals, overall finals and first-place tiebreak rounds together contribute at most 0.35 units per team/session. Longer schedules do not create an unlimited points or confidence bonus. Extra legitimate qualification appearances are retained within these limits.

Only diagonal variances carry into the next session. To limit overconfidence from discarded correlation, teamwork variance has a floor of 0.5 until three distinct partners have been observed, then 0.15. Repeated inseparable pairs remain provisional. These safeguards are heuristic; full persistent covariance and identifiable partner-graph diagnostics are future alternatives to evaluate in backtesting.

Final rank is not added as another rating term because finals score already supplies evidence. Official champions keep their titles even if a conservative performance estimate falls. The V5 candidate's champion floor is not copied into IQ: it can reward a strong selected partner twice. Any future achievement score must be separately named and reported.

Partner-adjusted consistency and evidence of independent skill

Revision 1.1 adds a 10% component for repeatable own contribution, reducing the direct teamwork weight from 65% to 55%. Driving and coding remain 17.5% each. This is a deliberate ranking policy, not an additional independent measurement: consistency and teamwork share qualification data. The uncertainty formula retains its conservative covariance bound.

Raw score variation alone is misleading. A team may maintain its own contribution while alliance totals vary with partner strength, or earn steady high totals because each partner is strong. The component therefore starts with the same frozen, bounded shared-score target used by the cooperative model.

For team i and its qualification partner j:

```
r = fitted shared target - prior_mean_i - prior_mean_j
share_i = prior_variance_i / (prior_variance_i + prior_variance_j)
a_i = prior_mean_i + share_i × r
```

The adjusted contribution a_i is a prior-regularized estimate of the team's own standardized contribution. When the partner already explains the observed score, there is no unexplained boost. Equal uncertainties split the residual equally. An uncertain partner weakens attribution. No estimate is a literal individual-point count.

Store only valid two-team qualification observations. Finals, scheduled solo matches, DQs, no-shows, practices and invalidated results cannot satisfy the consistency sample. Each stored row records its unique event/session/match key, date, partner identity, frozen partner mean/variance, adjusted contribution and capped teamwork evidence weight.

Use the most recent 24 observations within 90 days, ordered by timestamp and unique key. Their effective weights are:

```
w_i = teamwork_evidence_weight / (1 + partner_variance)
      × 2 ^ (-age_in_days / 45)
W = sum of effective weights
A = weighted mean of adjusted own contribution
S = weighted population standard deviation of adjusted own contribution
repeatability = 1 / (1 + S²)
```

Eligibility requires at least six positive-weight observations, three distinct partners and effective mass W >= 2. These thresholds prevent one partner, a short schedule, stale records or highly uncertain partners from producing a confident conclusion. A sample that fails these checks is labeled insufficient evidence; missing consistency is not redistributed into other weights. Combined rankings stay provisional until this requirement is met.

For an eligible window:

```
reliability = min(1, W / 6) × min(1, distinct_partners / 4)
supported_ability = clamp(A - 0.5 × S, -3, +3)
observation_variance = 0.75 / reliability
snapshot_mean = supported_ability / (1 + observation_variance)
snapshot_variance = max(0.15, observation_variance / (1 + observation_variance))
```

Recompute this snapshot from a fixed neutral prior. Do not feed yesterday's rolling snapshot back as a new independent observation; that would count overlapping matches repeatedly and shrink uncertainty artificially. On a session with new usable qualification observations and an eligible window, move the consistency component toward snapshot_mean by at most +/-0.6 units. Variance is the fixed-prior snapshot variance. Its evidence counter is the window's effective mass, not a lifetime cumulative count. All component changes, including consistency, remain within the combined +/-120 internal-point session cap.

If a new session lacks enough evidence, retain the previous component mean with inactivity uncertainty; report the newly insufficient window and exclude the team from confirmed combined rankings. Reading a leaderboard recalculates window eligibility for the requested date, so stale evidence cannot stay eligible just because the team has not attended another event. Finals and skills-only events do not update the consistency rating from old qualification rows.

Stable zero contribution has S=0 but A=0, so its consistency rating is neutral. Stable negative contribution stays negative. At the same positive A, larger S lowers supported ability. Uniformly weak output therefore does not gain points merely for being uniform.

Diagnostics use descriptive, explicitly uncertain signals:

Signal	Evidence requirement
Insufficient evidence	Window fails one or more coverage thresholds
Repeatable positive contribution	Eligible; A > 0.25 and S <= 0.5
Partner-explained upside	Eligible; mean frozen partner strength > 0.25 and A <= 0.25
Variable contribution	Eligible; S > 0.75, after the earlier signal checks
Mixed evidence	Eligible but none of the above thresholds apply

These signals are evaluated in the listed order. Partner-explained upside means that partners explain the above-baseline scoring evidence; it is not proof that the team was carried, contributed nothing, or violated a rule. Pair specialization, shared tasks, congestion, new robots and rapidly improving partners remain confounders. Poorly connected partner graphs and uncertain priors require caution even when the numeric thresholds pass.

Illustrative comparison with adjusted contribution values in standardized units:

Team pattern	A	S	Supported ability
Repeated own contribution [1,1,1,1,1,1]	1.00	0.00	1.00
Same average, variable [0,2,0,2,0,2]	1.00	1.00	0.50
Partner explains upside; own estimates all 0	0.00	0.00	0.00
Repeated own estimates all -1	-1.00	0.00	-1.00

The final snapshot also shrinks these targets according to partner coverage, age and reliability. These are examples, not measured real-team labels. A new game, model version or changed normalization requires rebuilding all consistency rows from frozen chronological priors.

Driving and autonomous coding skills

Process each mode independently, once per team/session:

```
performance = 0.70 × best official attempt + 0.30 × median official attempts
target_raw = (performance - mode.center) / mode.scale
target = prior_mean + clamp(target_raw - prior_mean, -3, +3)
reliability = min(1, observed_attempts / 3)
observation variance = 0.5 / reliability
gain = prior_variance / (prior_variance + observation variance)
delta = clamp(gain × (target - prior_mean), -0.6, +0.6)
new variance = max(0.1, (1 - gain) × prior_variance)
```

This rating statistic intentionally differs from the official peak-based skills standings. It rewards both demonstrated ceiling and repeatability. A scored zero, including a verified DQ zero, is an attempt. An unused, unplayed or missing attempt is not invented as a zero.

Many upstream records contain only the best score and perhaps an attempt count, not all attempt scores. For a `best-only` row use that peak, reliability 0.5, unknown median and unknown observed-attempt count. Do not fabricate a median from an aggregate or mix a best-only summary with individual attempts. A fourth or later played attempt needs explicit official verification, for example a permitted deciding run.

Missing driving or coding data leaves its prior mean and uncertainty in place. The other weights are not increased. Teams without evidence in both modes are provisional on the combined leaderboard; they can still appear in a clearly labeled provisional view.

`summarizeIQSkills` returns descriptive mode peaks and an official-style total, keeping missing mode values as `null` while treating a single submitted mode's absent counterpart as zero for the total. `bestIQSkillsSession` chooses a single session's combined total and flags unresolved ties. It never combines one session's driving peak with another session's coding peak.

Stop times are official tiebreak metadata, not bonus points or a reason to rescale a 60-second score into a hypothetical full-time total. The importer must preserve all available tie criteria and published ranks. These helpers do not claim to reproduce the entire official event/global tie resolver.

Public score, confidence, inactivity and resets

```
M = 0.55 × m_teamwork + 0.175 × m_driving
  + 0.175 × m_coding + 0.10 × m_consistency
mu = 1500 + 200 × M
RD = 200 × (0.55√v_teamwork + 0.175√v_driving
          + 0.175√v_coding + 0.10√v_consistency)
VIQCR = clamp(500 + 1.25 × (mu - 0.75 × RD - 1500), 0, 1000)
```

The RD formula sums weighted standard deviations. It is a conservative upper bound when cross-mode correlations are unknown, not a calibrated probability interval. Public IQ scores are not comparable with V5 VCR despite the familiar 0-1000 presentation. Keep full precision for ordering and round only for display; exact ties use a stable ID order, not an implied official tiebreak.

Every component change is bounded by +/-0.6, so a session can change the blended internal mean by at most 120 points. Public movement also reflects uncertainty and can differ from the mean change. Event-history entries store exact component changes and `internalChange`; calculate public scores from the corresponding state/as-of date, not by summing rounded history values.

Inactivity preserves the mean and increases variance by 0.0025 per day, up to 1.5. A separate `varianceAt` timestamp prevents applying the same elapsed days twice. Reading a rating does not mutate state. More uncertainty lowers the conservative public score without claiming the team became less capable.

A team is provisional until it has two teamwork sessions, three distinct observed partners, at least one recorded skills session in each solo mode, and currently eligible consistency evidence. Confirmed leaderboard output excludes provisional teams by default. Thresholds are initial coverage safeguards, not official participation requirements or proof of statistical calibration.

Between games, a weak prior may carry only when the importer confirms the same team identity: `new mean = clamp(0.25 × old mean, -0.5, +0.5)`, variance resets to 1, and evidence, partners, history and processed sessions reset. Consistency mean and rolling evidence reset to zero/empty rather than carrying a previous game's repeatability. Use a fresh state for a reused number, changed identity, program transfer or uncertain roster continuity. No V5-to-IQ carryover is supported.

Import and replay contract

Use official stable team IDs as strings, not team numbers. Season IDs are strings too. Teams have `id`, `number`, verified `grade` (`Elementary School`, `Middle School` or `Unknown`) and optionally `region`. Grade changes must come from the season's official classification.

Each normalized event/session supplies:

```
{
  id: 'official-event-id', sessionId: 'official-session-id-or-main',
  program: 'VIQRC', seasonId: 'official-iq-season-id-as-string',
  format: 'tournament', // or league-session / skills-only
  start: '2026-10-01T08:00:00Z', end: '2026-10-01T18:00:00Z',
  completed: true,
  coverage: { teamwork: true, skills: true },
  teams: [/* verified team identities/classifications */],
  teamwork: [{
    id: 'division-a/qualification/1', teamIds: ['team-id-1', 'team-id-2'],
```

```
    stage: 'qualification', // or finals / practice
    status: 'scored', // or unplayed / void / superseded
    score: 120, // verified single shared score
    noShowIds: [], dqIds: [], reliability: 1
  }],
  skills: [{
    id: 'skills-attempt-id', teamId: 'team-id-1', mode: 'coding',
    status: 'scored', score: 50, detail: 'attempts', // or best-only
    disqualified: false, stopTime: 0
  }],
  official: { /* published standings, finals scope/results, honors, sanctions and source provenance */ }
}
```

The completion and coverage booleans are assertions by a trusted importer, not independent verification by the mathematical engine. Fetch every relevant page, division and league session and resolve duplicate/corrected records before setting them. Successful confirmation that a mode was not offered can be a complete empty collection; an API failure cannot. Verify eligibility for all imported skills, including the rules for approved skills-only events. Never rate a canceled event or an unfinished league as a completed tournament.

`iqStationTeamIds` validates the two one-team station shape commonly used by IQ result APIs. The importer must separately verify the shared alliance score against official IQ data. Do not choose the maximum, sum station scores, compare stations as opponents, or assume a nonzero station always proves the shared total. Ambiguous station-score encodings must fail the import.

Result IDs must be unique across teamwork and skills within a session. Include division, round and instance/replay identity in normalized match IDs. Resolve a replacement replay with `superseded` on the old result. A first-place finals tiebreak is another legitimate scored result, subject to the shared finals budget.

Process sessions by `(end timestamp, JSON.stringify([event ID, session ID]))`, ascending. Equal-end ordering is deterministic. Overlapping events use that declared end-order approximation; expectations are frozen from the state at processing time, not a simulation of real-time access to results from another ongoing event. A duplicate event/session, out-of-order event, wrong program/season, missing status, invalid score or incomplete collection throws before changing state. Corrected upstream results require replay from before the correction.

At most 512 teams are supported per session in this candidate's dense covariance fit. Larger datasets need a reviewed sparse implementation or verified independent-division partitioning. When partitioning, retain a single shared per-team finals budget across any later overall final; do not reset that budget just to bypass this limit. Never silently truncate a large event.

The API returns `{replay, ledger, diagnostics}` and leaves input state intact. Replay contains a `Map`; persistence must serialize `teams` as entries and restore it to a `Map`. Persist model version, normalization version/signature, processed keys and last order together. Do not replace the V5 archive schema or cache keys with IQ output.

Implementation example

```
import { createIQReplay, processIQEvent, rankIQTeams } from './website/lib/viqcr.mjs';

let replay = createIQReplay(normalization.seasonId);
for (const event of verifiedChronologicalSessions) {
  ({ replay } = processIQEvent({ replay, event, normalization }));
}
const standings = rankIQTeams(replay, {
  asOf: '2026-11-01T18:00:00Z', grade: 'Elementary School',
```

```
});
```

`rankIQTeams` returns model order, not official qualification or skills order. A viewer should label it "VIQCR", show the model version, provide a provisional toggle, and display official records in separate fields. Unknown grade is never inferred from a school name. Regions and event prestige cannot change the same team's rating.

Validation and release criteria

Run from `website/`:

```
node --test tests/viqcr.test.mjs tests/vcr3.test.mjs tests/data-regressions.test.mjs
    tests/archive.test.mjs
```

The scenarios cover cooperative attribution, known partners, order symmetry, repeated schedules, bounded finals, solo appearances, zeros/missingness, DQs/no-shows, independent skills, peak/repeatability, partner-adjusted consistency, weak-output neutrality, partner uncertainty, rolling expiry, best-only data, league totals, qualification drops, grade filters, caps/audit arithmetic, inactivity, season reset, atomic replay and invalid inputs. V5 and archive checks protect existing behavior.

Before publishing live IQ rankings:

1. Implement and fixture-test official IQ data normalization, complete pagination, station shared-score extraction, eligibility, league session identity and all-division coverage. Preserve official ordering and tie metadata rather than guessing winners.

2. Fit a versioned same-game calibration cohort and replay a held-out chronological dataset. Evaluate future teamwork score error, future solo performance, component stability and rank correlation; do not fit and evaluate on the same events.

3. Compare against simple qualification-average and official-skills baselines. Audit partner-graph identifiability, repeated-pair behavior, missing modes, grade cohorts, region connectivity, score ceilings, early-season coding zeroes and game-version changes.

4. Validate whether additive partner contributions adequately represent the game. Shared tasks, congestion, role specialization and score saturation can create pair synergy that cannot be identified from shared scores alone. Do not label the fitted contribution as ground-truth individual points.

5. Tune weights, caps, uncertainty, recency and provisional thresholds using held-out results. Run sensitivity checks for participation opportunities and best-only feeds. Consider persistent covariance or explicit synergy only if evidence supports the complexity.

6. Add IQ-specific API routes, archive envelopes, cache namespaces and UI program/season choices. Display coverage and provisional status. Deploy only after official import and historical validation pass.

This candidate covers how each competition aspect enters the model or remains an official record. It does not claim that all aspects of student achievement can be recovered from competition scores, or that scenario tests establish real-world ranking accuracy.

Official source references

Rules and judging sources reviewed for the competition structure. Statistical weights, confidence policies and consistency thresholds are Vex-Rank model choices, not official VEX ranking rules.

Level Up v2.1 - manual version and changelog

https://manuals.vex.com/iq/level-up/v2.1/prefix/

Tournament rules - qualifications, finals, skills and leagues

https://manuals.vex.com/iq/level-up/v2.1/event/

Game rules - disqualifications, robot handling and driver switch

https://manuals.vex.com/iq/level-up/v2.1/game/

Robot Skills rules

https://manuals.vex.com/iq/level-up/v2.1/skills/

Current IQ game overview

https://www.vexrobotics.com/iq/competition/viqc-current-game

Official IQ Level Up Q&A

https://events.vex.com/VIQRC/2026-2027/QA?filter=new_question

Guide to Judging: Awards

https://kb.roboticseducation.org/hc/en-us/articles/4912455338391-Robotics-Guide-to-Judging-Awards